INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

TESTER, DENSITY AND MOISTURE (SOIL AND ASPHALT), NUCLEAR METHOD

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1 SCOPE This commercial item description covers a one-piece (radioactive) source density and moisture tester. The tester is used for measurement of density and moisture content of soils, base materials, and density of asphaltic concrete

2. SALIENT CHARACTERISTICS. The tester shall be a single unit radiographic device consisting of radioactive sources, radioactive detectors, a readout device, memory data storage, an internal power supply, and housing The tester shall be capable of testing for density and moisture by backscatter, direct transmission, and (optional) air gap methods. Depth of measurement shall be up to 8 inches in 2-inch increments

2 1 Radioactive sources. The sources shall be encapsulated and sealed.

2.2 Radioactive detectors The detectors shall be any suitable type of radioactive detectors

2.3 Readout device. The readout device shall contain a power supply, counting circuits, timing circuits, a data display, and related electronic components. The display shall be direct readout in wet density, dry density, and moisture content in pounds per cubic foot (pcf) with touch button control for conversion to metric units. The display shall clearly indicate the quantity and units of measure, in any ambient light.

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2 4 Memory data storage The tester shall have a memory data storage of a minimum of 120 test records with the capability to transfer to an electronic printer or computer

2.5 Power supply. The power supply shall consist of a battery pack capable of a minimum of 8 hours of normal consecutive uses with measurements in the normal 60-second operational mode prior to a recharge or battery pack replacement It shall be comprised of batteries that minimize the promotion of unit corrosion, if left installed for extended periods

2 6 Battery voltage indicator An indicator or device shall provide a clear distinction between when the battery voltage is adequate for testing and low battery reserve When low battery reserve is indicated, there shall be at least 2 hours of consecutive uses prior to battery recharge or replacement

2 7 Housing The power supply source, detectors, readout device, etc., shall be in a housing of rugged construction that is moisture and dust proof The housing shall conform to the U S. Nuclear Regulatory Commission requirements in Title 10 of the Code of Federal Regulations. Radiation labels required by 2 8 5 for the transportation case shall also be applied to the housing.

2 8 Tools and accessories The tester shall be furnished with a reference standard, site-preparation devices, technical publications and, when required, a battery charger A driver/puller device(s), a transport case and all other tools necessary for operation of the tester shall also be supplied.

2.8.1 Reference standard. The reference standard shall be a combination type provided for the purpose of checking equipment operation, background count, and to establish conditions for a reproducible density and moisture count rate

2 8 2 Site-preparation devices The devices shall provide for the following tests to comply with performance requirements specified in 2 9

2 8 2 1 Backscatter. Site-preparation devices for the backscatter test method shall include a flat plate straightedge or other suitable tools, to be used to level the test site to the required smoothness

2.8.2.2 Direct transmission. In addition to the tools specified in 2 8.2 1, site-preparation devices for the direct transmission test method shall include an auger, pin or other hole-forming device having a nominal diameter equal to or up to 1/8 inch larger than the probe, and a guide to ensure that the hole is perpendicular to the test site surface

2 8 2 3 Air gap (optional) In addition to the tools specified in 2.8.2 1, and in 2.8.2.2, site-preparation devices for the (optional) air gap test method shall include a cradle or spacers, to support the gage at the optimum air gap above the material being tested

2.8 3 Battery charger When one is required, the built-in automatic charger shall be provided with quick-disconnect plug-in type connectors The tester shall incorporate a battery overcharge preventive feature The charger shall operate from either 12 volts dc or 110 volts, 60 hertz, ac and shall be designed to protect the voltage regulator from damage should the charging battery be connected improperly

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2 8 4 Technical publication. Technical publications shall include supplier's standard operating, maintenance, safety, and parts manuals.

2.8.5 Transport case The transport case shall contain the tester, all tools, accessories and technical publications The case shall conform to the U.S. Department of Transportation requirements in Title 49 of the Code of Federal Regulations. The exterior of the case shall contain all labels as required by the regulations

2.9 Performance. The tester shall meet the performance requirements of ASTM Standards D 2922, D 2950, and D 3017. The tester shall also meet the following requirements

2 9 1 Direct transmission

Range of calibration	80 to 160 pounds per cubic feet (pcf)
Precision (6-inch depth) (at 120 pcf)	± 0.25 pcf
Chemical error (at 120 pcf)	± 1 50 pcf
Surface roughness error (0 05 in)	- 0 90 pcf

2 9.2 Backscatter

Range of calibration	80 to 160 pcf
Precision (6-inch depth) (at 120 pcf)	± 0 50 pcf
Chemical error (at 120 pcf)	± 2.50 pcf
Surface roughness error (0 05 in)	- 4.00 pcf

2 9.3 Moisture

Range of calibration	0 to 40 pcf
Precision (at 10 pcf)	± 0 25 pcf

2.10 Temperature conditions. The tester shall operate as specified herein in any ambient temperature from 32 to 140 F. The tester shall operate as specified herein on hot materials (asphalt) at temperatures up to 345 F without the use of additional insulating material between the instrument and the surface.

2 11 Weight. The tester shall weigh not more than 42 pounds. The tester and case (with tools and accessories) shall weigh not more than 116 pounds.

2.12 Workmanship All parts shall be free from rust, tool marks, pits and other injurious defects External surfaces shall be free of burrs, sharp edges and corners except where sharp edges or corners are required or where they will not be detrimental to safety Electronic components shall be protected against adverse effects of moisture and fungus growth Each unit shall undergo burn-in of sufficient duration to ensure sustained precision over the life of the tester

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3 REGULATORY REQUIREMENTS The supplier shall comply with the following for radiation and recovered materials

3.1 Licensing and control of radiation source. The contractor shall certify that the tester has met the product safety requirements of the U.S Nuclear Regulatory Commission for products containing radioactive sources, and that the necessary licensing data has been provided. The contracting officer will furnish the contractor a copy of the license(s) prior to shipment date required

3 2 Recovered materials The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23 403 of the Federal Acquisition Regulation (FAR)

3 3 Radiation markings and labels. The contractor shall ensure that proper radiation markings and labels are affixed to all parts of components containing radioactive material

4 QUALITY ASSURANCE PROVISIONS The supplier shall comply with the following for certification and acceptability

4 1 Contractor certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

4 2 Market acceptability The tester offered shall be the latest model of the standard product of the supplier and shall have been used in the field for at least 1 year by professional organizations such as State Highway Departments

5 PACKAGING Preservation, packing, and marking shall be as specified in the contract or order

6 NOTES

6 1 The procuring agency should specify the level of preservation, packing, and marking required The procuring agency may also need to specify inclusion of a gamma detection meter (monitor) for the users radiation safety officer since it may be a requirement by some local regulatory agencies

6 2 When this commercial item description is used for procurement, the contractor certification clause must appear in the solicitation

6.3 Safety conformance The Army procuring agency must specify that the manufacturer provide.

a NRC Sealed Source and DOT 7A Specification Certification,
b Caution Radioactive Material Signs, Section 206 (Energy Reorganization Act of 1974) and the latest version of NRC Form 3 with each tester,
c Tester Transport Cases that are Moisture Resistant and the Case External Hardware Corrosion Resistant.

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d. Testers that have a different Auditory Sound for when the Testers are in Operational Mode,
e Testers in which Serviceable Subassemblies can be removable without exposing the Source or disturbing the Shielding Assembly, and
f Labels on the Tester Housing and the Transport Case that specifies the Serial and Model Number, Date of Source Manufacture, Type of Radioactive Sources, and Amount of Radioactive Material with a radiation caution symbol.

6 4 Source of documents Referenced documents can be obtained from the following sources

6 4 1 ASTM D 2922, D 2950 and D 3017 may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103

6.4.2 Federal Acquisition Regulation (FAR) and Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S Government Printing Office, Washington, DC 20402

6 5 Products known to meet the CID (for information only)

CPN 2830 HOWE ROAD MARTINEZ, CA 94553	MODEL MC-3 & MODEL MC-1DR
HUMBOLDT MFG CO 7300 W. AGATITE AVENUE NORRIDGE, IL 60656	MODEL HS-5001-C
SEAMAN NUCLEAR CORPORATION 7315 S FIRST STREET OAK CREEK, WI 53154	MODEL C-75
TROXLER ELECTRONIC LABORATORIES, INC P O BOX 12057 RESEARCH TRIANGLE PARK, NC 27709	MODEL 3440

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